

3.1 Exploratory Data Analysis

Récupération, analyse et visualisation de données numériques

UniGe – A.A. 2024/2025

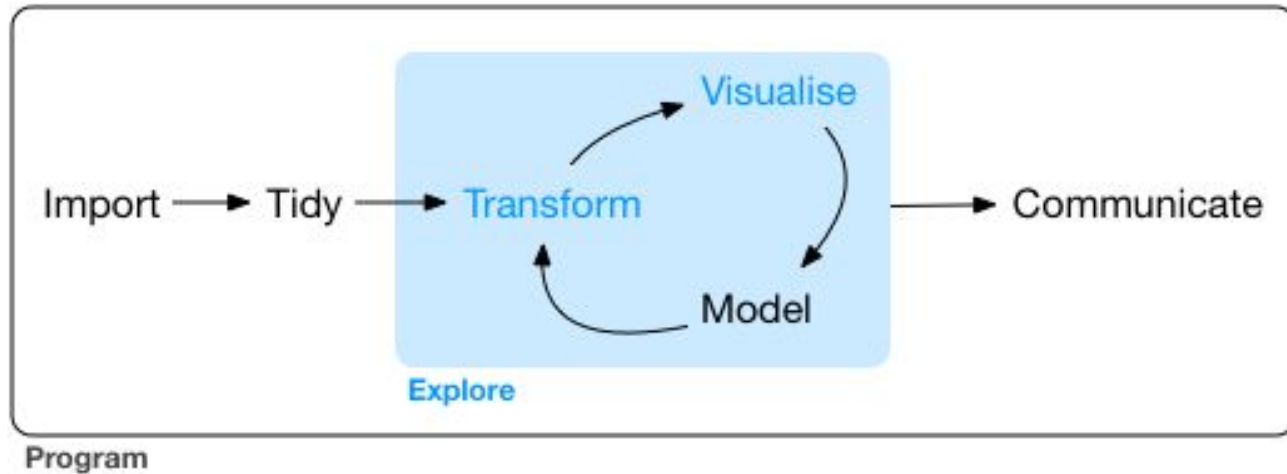
Exploratory data analysis

We want to use visualisation and transformation to explore your data in a systematic way, a task that statisticians call **exploratory data analysis**. Exploratory data analysis is the **iterative process** of:

1. Generating questions about your data.
2. Searching for answers by visualising, transforming, and modelling your data.
3. Using what you learn to refine your questions and/or generate new questions.

Exploratory data analysis is an important part of any data analysis, even if the questions are handed to you on a platter, because you always need to investigate the quality of your data. **Data cleaning** is just one application of it: you ask questions about whether your data meets your expectations or not.

Exploratory data analysis

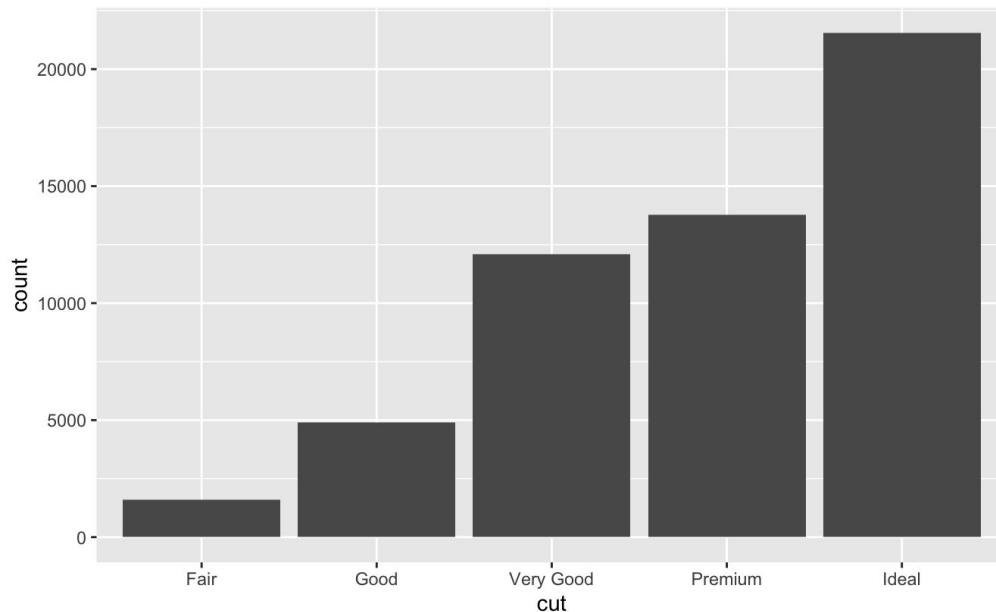


Today

- Basic plots: histograms, scatter plots, bar plots and box plots
- Two important distributions: normal and long-tail
- Descriptive statistics
- Outliers
- ~~• Measuring change~~
- ~~• Measuring co-variation~~

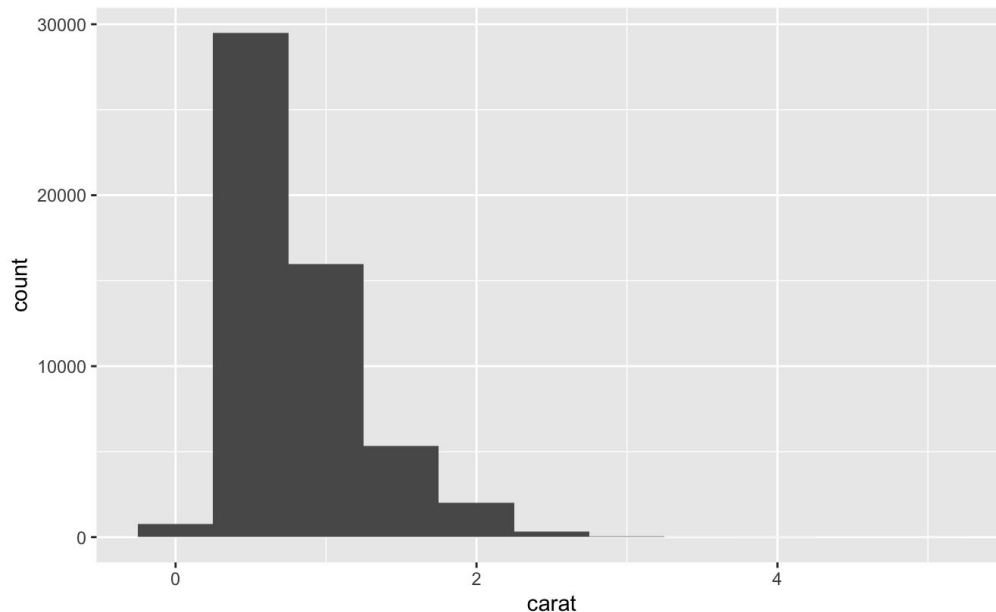
Visualizing variation in data

Categorical variables: bar plots



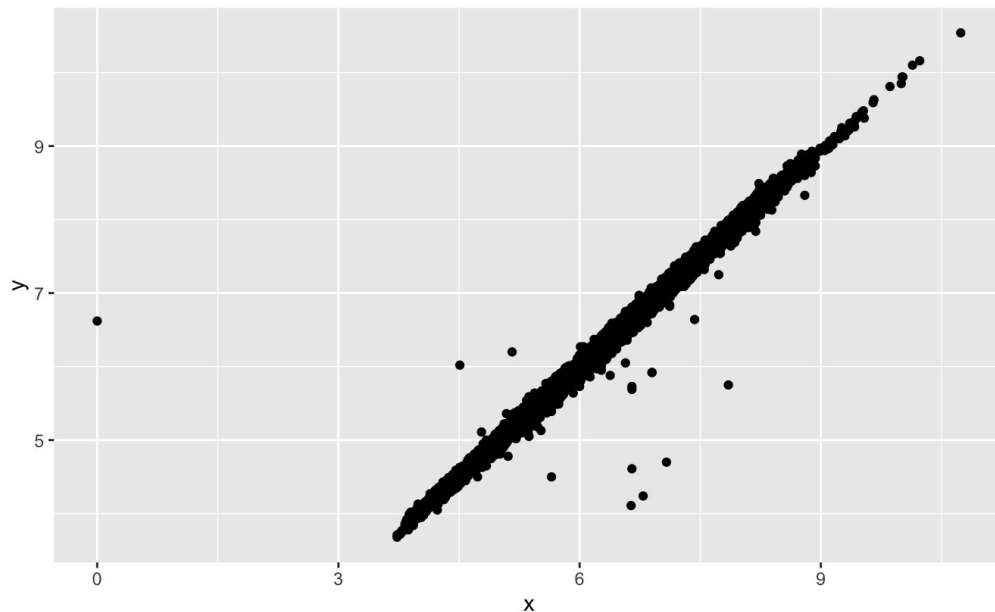
Visualizing variation in data

Continuous variables: histograms

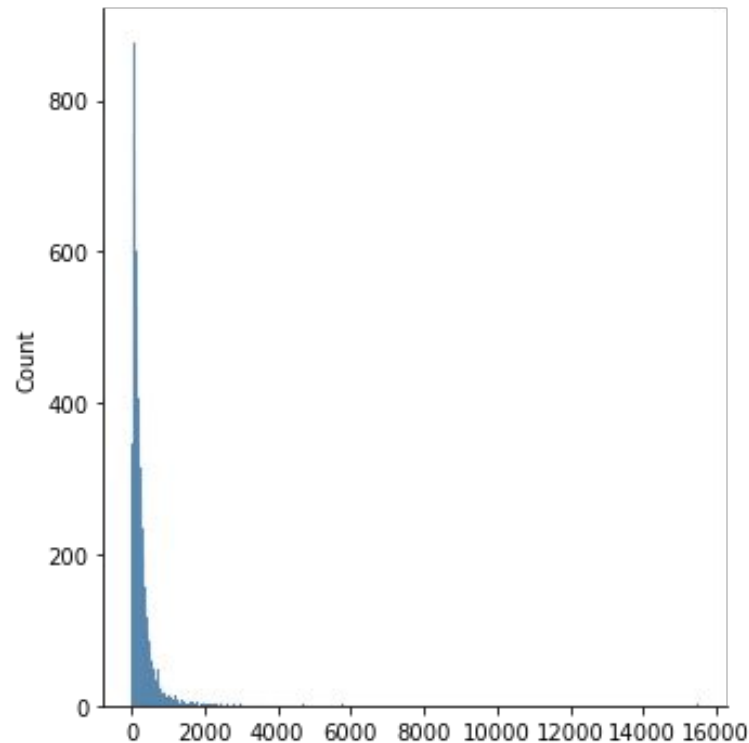
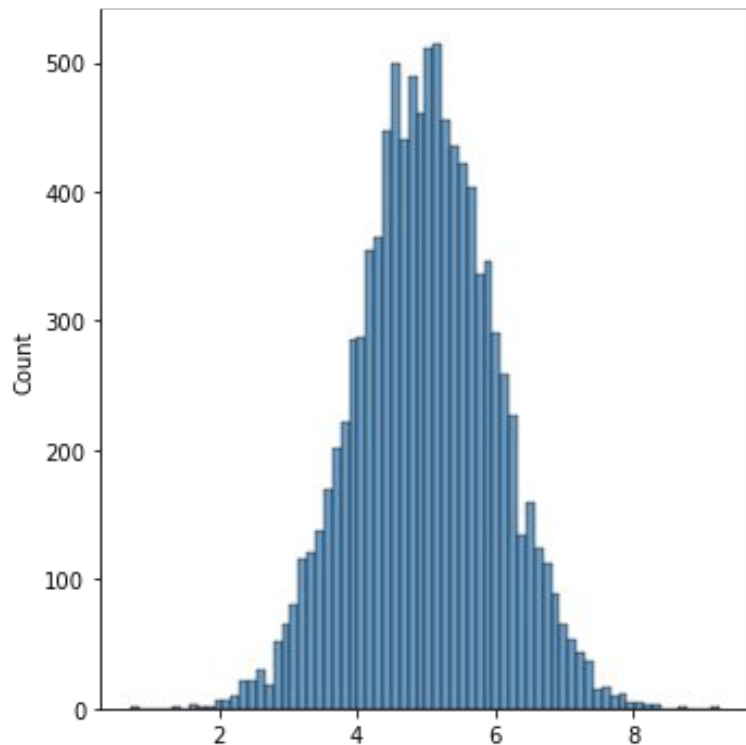


Visualizing variation in data

Comparing two continuous variables: scatter plots



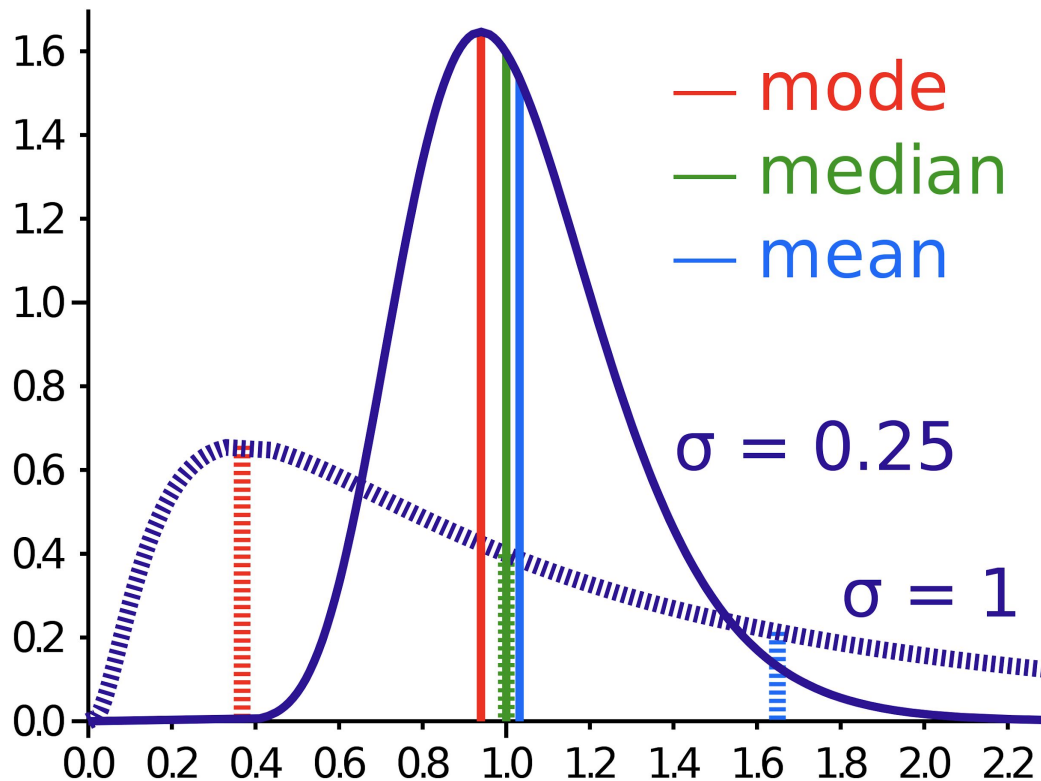
Distributions: normal vs long-tail



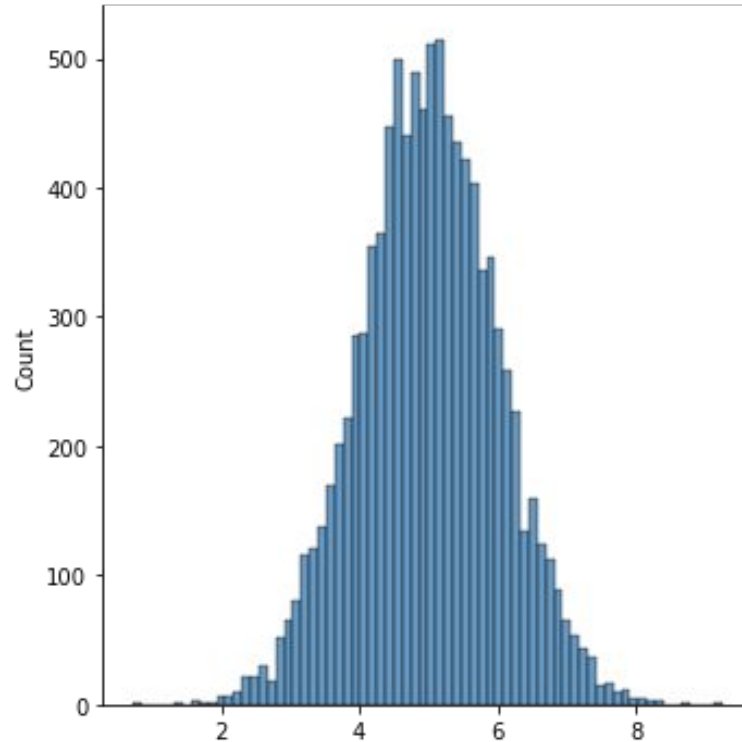
Descriptive statistics

- **Mean:** average value
- **Mode:** most frequent value
- **Median:** value such that 50% of data points are below and 50% above it
- **Quartile:**
 - 1st) value such that 25% of data points are below and 75% above it
 - 2nd) the median
 - 3rd) value such that 75% of data points are below and 25% above it
- Other useful stats: minimum, maximum, **standard deviation** (a measure of spread)

Descriptive statistics



The normal distribution (bell curve, Gaussian curve)



Properties of the normal distribution / bell curve

The distribution that occurs naturally in many situations.

E.g., **the bulk of students will score the average (C)**, while smaller numbers of students will score a B or D. An even smaller percentage of students score an F or an A. This creates a distribution that resembles a bell.

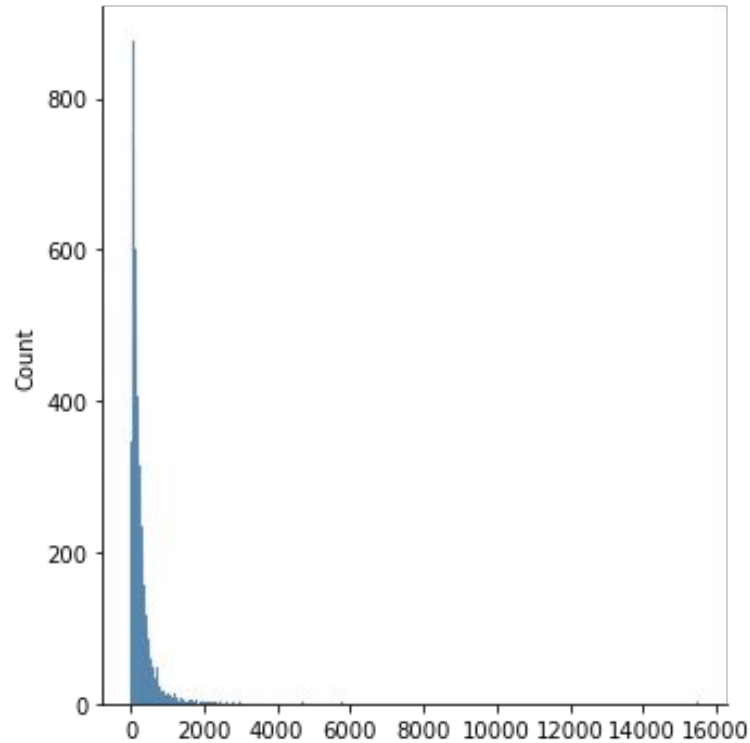
Other examples:

- Heights of people.
- Measurement errors.
- Blood pressure.
- Points on a test.
- IQ scores.

Properties of a normal distribution

- The curve is symmetric at the center (i.e., around the mean).
- Half of the values are to the left of center and half the values are to the right.
- The mean, mode and median are all equal or very close.

Long-tail distributions / Zipf distribution



Examples of long tail distributions

Many socioeconomic and cultural phenomena take long-tailed distributions:

- city population sizes
- occurrences of natural resources (e.g., size of reserves in a certain geological region)
- stock price fluctuations
- size of companies
- word frequencies
- named entities

Descriptive statistics

4 Probability and Statistics

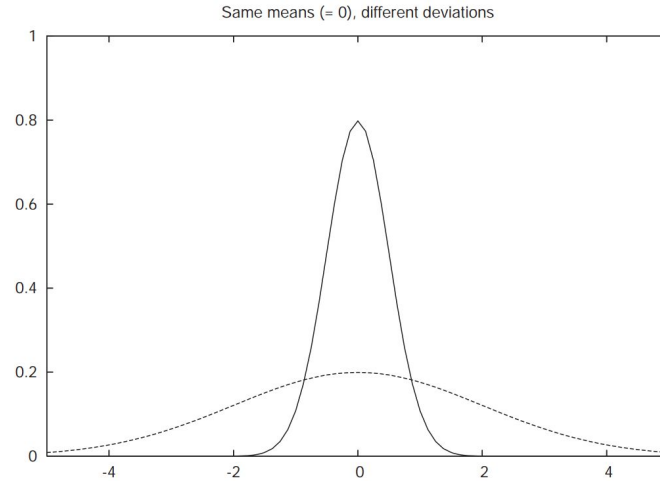
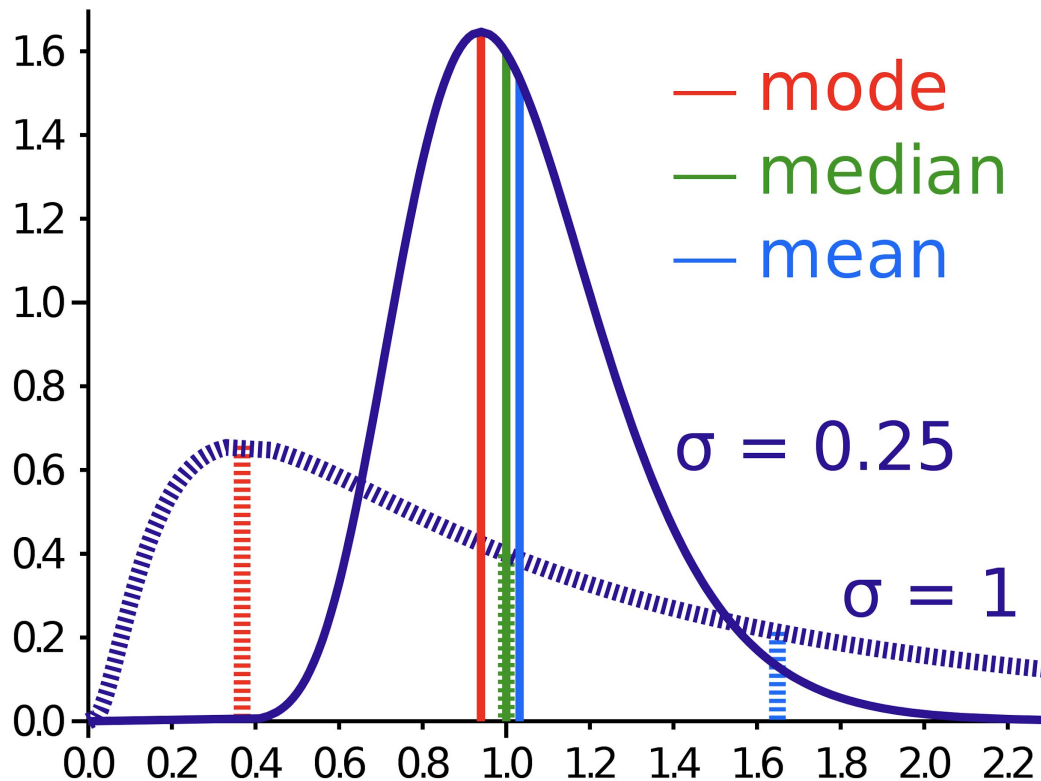


Figure 4.14: Two different bell curves

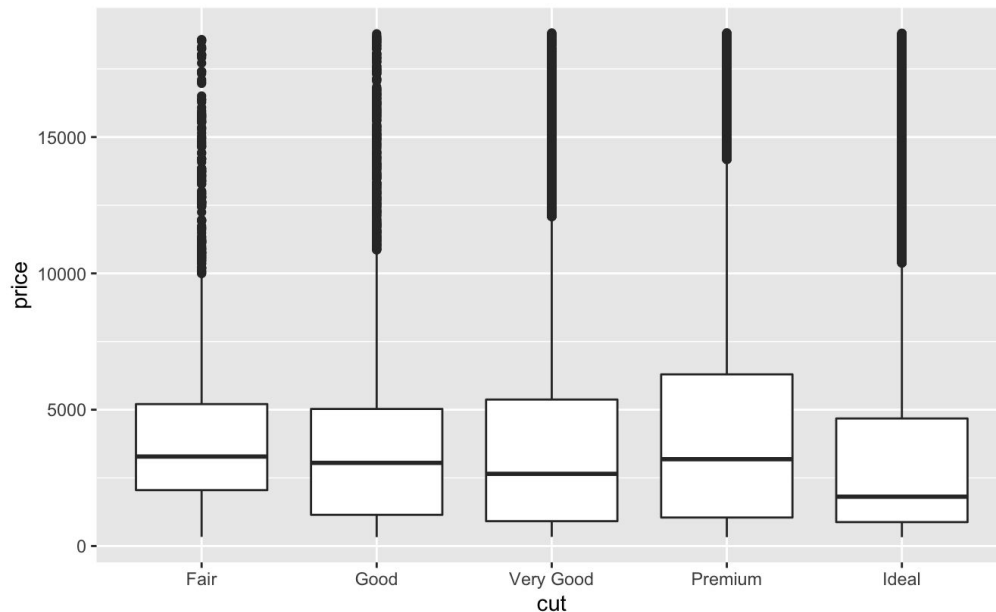
P. Juola and S. Ramsay, [Probability and Statistics of the book Six Septembers: Mathematics for the Humanist.](#)

Descriptive statistics



Visualizing variation in data

Exploring the distribution of a continuous variable and expose outliers: box plots

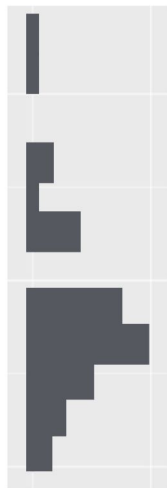


Visualizing variation in data

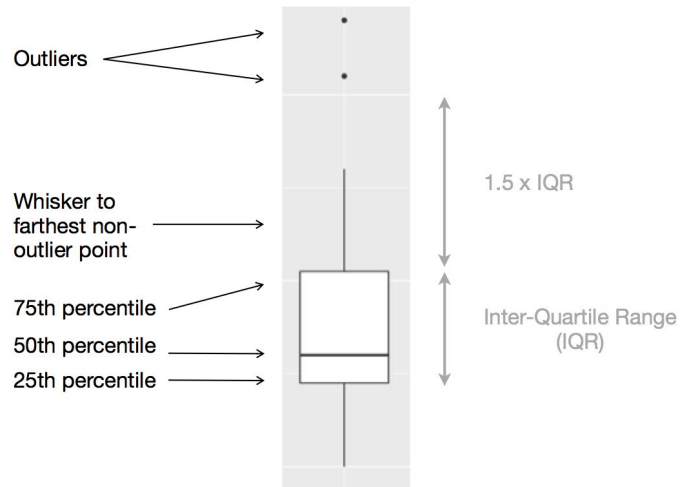
The actual values in a distribution



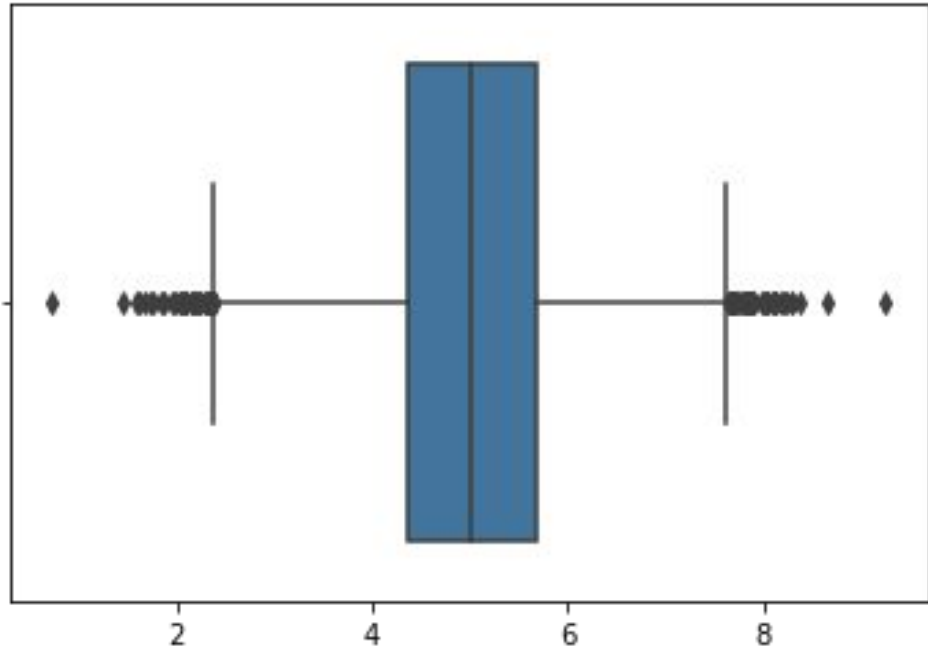
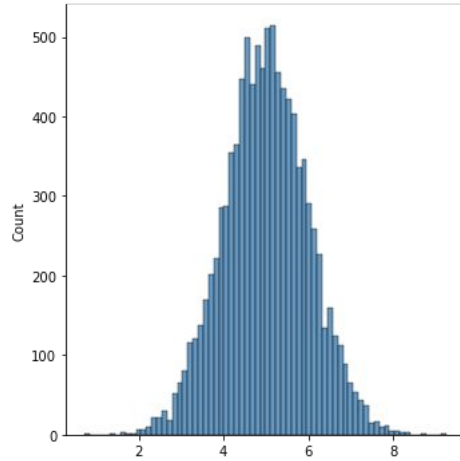
How a histogram would display the values (rotated)



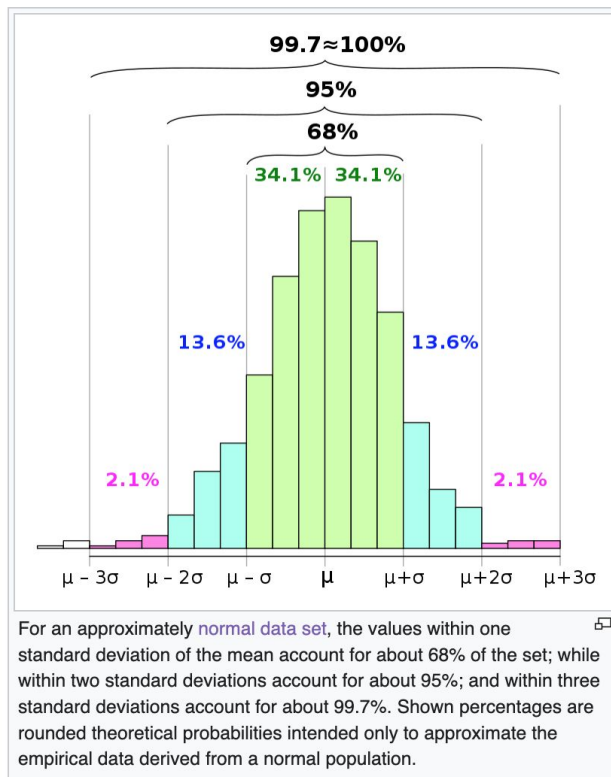
How a boxplot would display the values



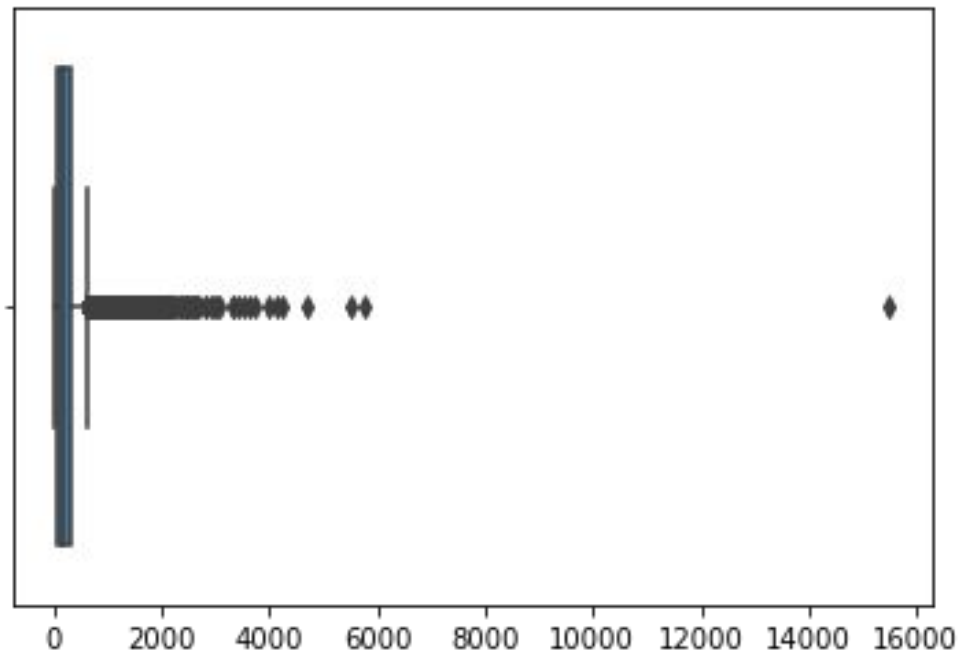
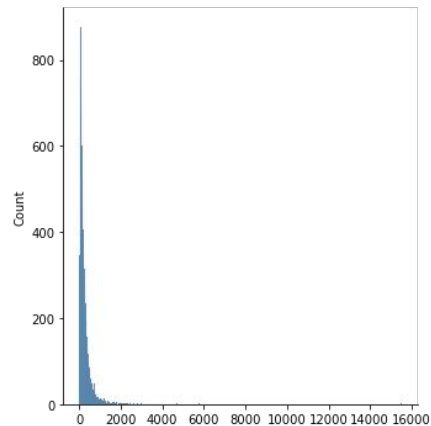
The normal distribution



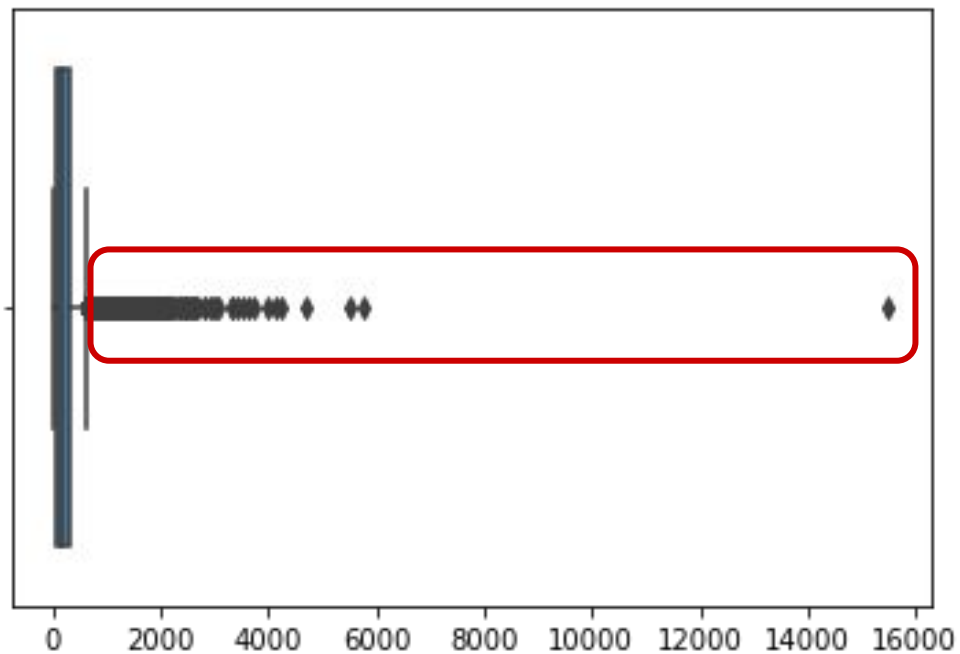
The normal distribution



Long-tail distributions



Outliers



Measuring change

An effective measure of change are % increases (or decreases) over a certain time.

The calculation is very simple: $((\text{new_value} - \text{old_value}) / \text{old_value}) * 100$

Example:

- house price index 2019: 500
- house price index 2020: 550
- house price index 2021: 490

Change 2019 to 2020: +10%; change 2020 to 2021: -10.9%

Measuring co-variation

How do two variables change together, when considering the same observations?

Covariance is a linear measure of such variation:

$$\text{cov}(X, Y) = \text{E} [(X - \text{E}[X])(Y - \text{E}[Y])], \quad (\text{Eq.1})$$

Q&A